Navigation Systems

Estimating Where You Are Without GPS

PHASE 8 — RESEARCH-GRADE NOTES

Trinetra-AI

An Intelligent Multi-Sensor Fusion Framework for GPS-Denied Navigation

> Navigation is the discipline of answering *where am I, which way am I facing, and how do I get there* — continuously and reliably. This phase assembles the navigation *systems* that turn fused sensor estimates into position, velocity, orientation, heading, and trajectory: GPS/GNSS, inertial navigation (INS), dead reckoning, magnetic and map-matching navigation, localization, and path planning — always from the standpoint of **operating when the satellites are gone**. Intuition first, then the mathematics, then derivations, then code, then the exact place in Trinetra-AI.

Traditional → satellite → inertial → magnetic & map-matching → AI-enhanced navigation.

Contents

Part I — Foundations

1 Introduction to Navigation

1.1 What is navigation?

Navigation is the process of determining and controlling a craft's *position*, *velocity*, and *orientation* over time, and planning/following a route to a destination. The estimation part is often abbreviated **PNT** — Position, Navigation, and Timing. Formally it is the online estimation of the pose trajectory $\{\mathbf{x}_t\}$ from whatever measurements are available.

Intuition: a sailor without instruments still navigates — tracking speed, heading, and elapsed time to reason about position (dead reckoning), and correcting against absolute references (stars, landmarks). Every modern system does the same: a smooth but drifting internal estimate, periodically anchored by an absolute fix. Trinetra-AI's absolute anchor, when GPS is gone, is the magnetic field and maps.

1.2 History and evolution

Era	Method	Principle
Ancient	celestial, compass, dead reckoning	stars, magnetism, speed×time
20th c.	radio (LORAN, VOR), inertial	time-of-arrival; integrate motion
Satellite	GPS/GNSS	trilateration from orbiting clocks
Modern	INS/GNSS fusion, SLAM	optimal estimation, mapping
Frontier	magnetic-anomaly & quantum nav	Earth's field as a passive, unjammable map

1.3 Why navigation is difficult

Every absolute reference can vanish (GPS indoors, jamming), and every self-contained method drifts (inertial integration). The core tension is between *smooth-but-drifting* (inertial) and *absolute-but-sparse/fragile* (GPS, magnetic map) sources — resolved only by fusion. In GPS-denied settings the absolute anchor is scarce, so bounding drift becomes the central engineering problem.

1.4 Applications

Domain	Navigation emphasis (GPS-denied relevance)
Robotics	SLAM, wheel+IMU odometry indoors (no GPS)
Autonomous vehicles	GNSS+INS+map; urban-canyon multipath & tunnels
Aircraft / drones	INS/GNSS; magnetic-anomaly aiding when jammed
Spacecraft (ISRO/NASA)	star-tracker + IMU; no GPS beyond MEO
Submarines	INS + gravimetry/geomagnetics (no GPS underwater)
Defense / missiles	jam/spoof-resistant terrain & magnetic navigation

Trinetra-AI is a navigation system whose defining constraint is *GPS denial*. Phases 4–7 built the sensing and fusion machinery; this phase organises them into the actual navigation modes — inertial dead reckoning bounded by magnetic-map matching and barometric altitude — that keep a position estimate alive when satellites cannot.

2 Coordinate Systems

Navigation juggles several frames: **geodetic** (latitude, longitude, altitude on the ellipsoid); **ECEF** (Earth-Centred Earth-Fixed Cartesian); **ENU/NED** (local tangent East-North-Up / North-East-Down); **body** (fixed to the vehicle); **navigation** (local level); and **map** (the reference frame of the anomaly/road map). Fusion requires transforming every measurement into one common frame.

Geodetic (φ, λ, h) to ECEF (ellipsoid semi-major a, eccentricity^2 e^2, $N = a/\sqrt{1 - e^2 \sin^2 \varphi}$):

$$x = (N + h)\cos\varphi\cos\lambda, \quad y = (N + h)\cos\varphi\sin\lambda, \quad z = \big(N(1 - e^2) + h\big)\sin\varphi.$$

ECEF to local ENU about a reference point uses the rotation $\mathbf{p}_{\mathrm{ENU}} = \mathbf{R}(\varphi_0, \lambda_0)\,(\mathbf{p}_{\mathrm{ECEF}} - \mathbf{p}_{\mathrm{ref}})$. NED is ENU with axes reordered/signed: $[E, N, U] \to [N, E, -U]$ — a frequent bug source.

```
import numpy as np
def geodetic_to_ecef(lat, lon, h, a=6378137.0, e2=6.69437999e-3):
    s=np.sin(lat); N=a/np.sqrt(1-e2*s*s)
    return np.array([(N+h)*np.cos(lat)*np.cos(lon),
                     (N+h)*np.cos(lat)*np.sin(lon), (N*(1-e2)+h)*s])
def ecef_to_enu(p, ref_llh, ref_ecef):
    lat,lon,_ = ref_llh; d = p - ref_ecef
    R = np.array([[-np.sin(lon), np.cos(lon), 0],
                  [-np.sin(lat)*np.cos(lon), -np.sin(lat)*np.sin(lon), np.cos(lat)],
                  [ np.cos(lat)*np.cos(lon), np.cos(lat)*np.sin(lon), np.sin(lat)]])
    return R @ d
```

Part II — Satellite Navigation

3 GPS

Even a GPS-denied system must understand GPS deeply — it is the anchor used *when available* and the thing whose absence defines the problem.

3.1 Architecture: the three segments

GPS has three segments. The **space segment** (∼31 satellites in ∼20 200 km MEO orbits, each carrying **atomic clocks**) broadcasts timing and ephemeris. The **control segment** (ground stations) tracks and corrects the satellites. The **user segment** (receivers) computes position from the signals.

3.2 Positioning: pseudorange trilateration

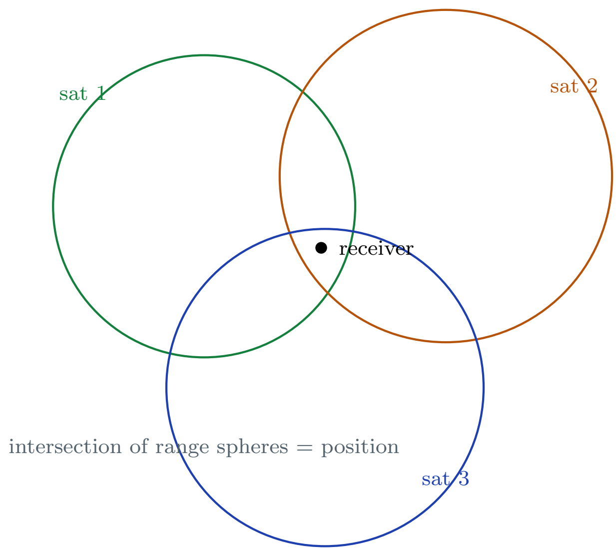
The receiver measures signal travel time, but its own clock is imperfect, so it obtains a **pseudorange**

$$\rho_i = \|\mathbf{p} - \mathbf{s}_i\| + c\,\delta t_u + \epsilon_i,$$

with $\mathbf{p}$ the unknown receiver position, $\mathbf{s}_i$ satellite i's position, δt_u the receiver clock bias, c the speed of light. Four unknowns $(x, y, z, \delta t_u)$ require ≥ 4 satellites. Linearising about a guess $\mathbf{p}_0$ gives the iterative least-squares update

$$\Delta\boldsymbol{\rho} = \mathbf{H}\begin{bmatrix}\Delta\mathbf{p} \\ c\,\Delta t_u\end{bmatrix}, \qquad \mathbf{H}_i = [\,-\hat{\mathbf{u}}_i^{\mathsf{T}} \;\; 1\,],$$

$\hat{\mathbf{u}}_i$ = unit vector to satellite i. Satellite geometry sets the error amplification (**DOP**, dilution of precision). **Carrier-phase** measurements (counting signal wavelengths) enable cm-level RTK positioning but require integer-ambiguity resolution.



3.3 Error sources — and why GPS fails

Error	Cause	Effect
Multipath	reflections off buildings/terrain	non-Gaussian metre-level jumps
Atmospheric delay	iono/troposphere refraction	ranging bias (partly correctable)
Satellite clock-/ephemeris	imperfect time/orbit	ranging bias
Receiver noise	thermal, tracking	random error
Urban canyon	blocked/reflected signals	high DOP, multipath
Indoor / underwater	signal cannot penetrate	**no fix at all**
Jamming / spoofing	interference / deception	denial or false position

```
import numpy as np
def solve_gps(sats, rhos, x0=np.zeros(4), iters=8):
    x = x0.astype(float) # [x,y,z, c*dt]
    for _ in range(iters):
        d = sats - x[:3]; r = np.linalg.norm(d, axis=1)
        pred = r + x[3]; H = np.hstack([-d/r[:,None], np.ones((len(r),1))])
        x = x + np.linalg.lstsq(H, rhos - pred, rcond=None)[0] # Gauss-Newton
    return x
```

GPS in Trinetra-AI: when available, a gated GPS fix (chi-square rejection of multipath) resets accumulated inertial drift — the ideal absolute anchor. The entire project exists for *when it is not* available: the fusion core then falls back to inertial dead reckoning bounded by magnetic-map matching and barometric altitude.

4 GNSS

GNSS (Global Navigation Satellite System) is the umbrella term for *all* satellite-positioning constellations; **GPS** is just the U.S. one. Using several constellations at once (**multi-GNSS**) gives more visible satellites, better geometry (lower DOP), and redundancy.

System	Owner	Notes
GPS	USA	original; global; L1/L2/L5
GLONASS	Russia	global; FDMA/CDMA
Galileo	EU	global; high-accuracy civil
BeiDou	China	global (3rd gen)
NavIC (IRNSS)	India (ISRO)	*regional* (India + 1500 km); L5/S, now L1
QZSS	Japan	regional augmentation over Asia-Pacific
SBAS	regional	wide-area corrections (WAAS, EGNOS, GAGAN)

NavIC is a live cautionary tale for this project. India's regional system nominally needs seven satellites, but by early 2026 multiple *atomic-clock failures* across the first-generation IRNSS satellites and the NVS-02 satellite's failure to reach its operational orbit left only about three–four fully operational — below the threshold for robust service. Replacements (NVS-03/04/05, carrying multiple indigenous rubidium clocks and the new smartphone-compatible L1 signal) are planned to restore it. The lesson is exactly Trinetra-AI's thesis: *even a nation's own GNSS is fragile* — clocks fail, launches miss orbit, signals get jammed — so a serious navigation system cannot depend on satellites alone. GPS-denial is not a corner case; it is the design centre.

Why GNSS still fails in GPS-denied environments. Adding constellations helps geometry and availability, but every GNSS shares the same physics: weak signals from distant satellites that *cannot penetrate* indoors/underground/underwater, *reflect* in urban canyons (multipath), and can be *jammed or spoofed*. Multi-GNSS mitigates outages; it does not solve denial. That is why inertial, magnetic, and map-based navigation are indispensable.

Part III — Self-Contained Navigation

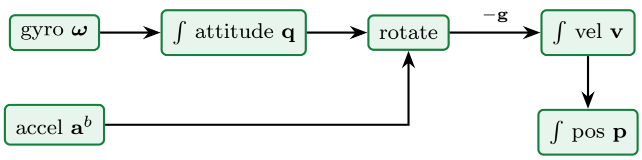
5 INS (Inertial Navigation System)

An **Inertial Navigation System** computes position, velocity, and attitude purely from an **IMU** (accelerometer + gyroscope), by integrating motion forward from a known initial state — *self-contained*, needing no external signal. A *strapdown* INS (sensors fixed to the body, no gimbals) is the modern norm.

The INS integrates three coupled equations — attitude, velocity, position:

$$\dot{\mathbf{q}} = \tfrac{1}{2}\mathbf{q} \otimes [0, \boldsymbol{\omega}], \qquad \dot{\mathbf{v}}^n = \mathbf{R}_b^n(\mathbf{q})\,\mathbf{a}^b - \mathbf{g}, \qquad \dot{\mathbf{p}}^n = \mathbf{v}^n,$$

i.e. integrate gyro rate to attitude, rotate specific force to the navigation frame and subtract gravity to get acceleration, integrate to velocity, integrate again to position.



Because the INS integrates, errors accumulate without bound: a gyro bias b_g gives heading error $\propto t$, mis-rotating gravity and yielding velocity error $\propto t^2$ and position error $\propto t^3$; an accelerometer bias b_a gives position error $\frac{1}{2}b_a t^2$. These polynomial laws set how long an INS can coast before it *must* be aided.

Grade	**Gyro bias**	**Unaided drift**
Consumer MEMS	10–100°/hr	metres in seconds
Tactical	1–10°/hr	useful for minutes
Navigation (FOG/RLG)	< 0.01°/hr	km-class over hours

Why INS is essential in GPS-denied navigation: it is the only *fully self-contained*, high-rate, unjammable source of motion — it works everywhere, indoors, underwater, under jamming. Its fatal flaw (unbounded drift) is precisely what the rest of Trinetra-AI exists to bound. The INS provides the continuous backbone; magnetic-map matching, barometer, and (when present) GPS supply the absolute anchors that reset its drift.

6 Dead Reckoning

Dead reckoning (DR) estimates the current pose by advancing a known pose using measured motion (heading + speed, or full inertial integration) with *no* external position fix. It is the algorithmic core of an unaided INS and of wheel-odometry systems.

Given speed v_t and heading ψ_t, position advances as

$$x_{t+1} = x_t + v_t \cos\psi_t\,\Delta t, \qquad y_{t+1} = y_t + v_t \sin\psi_t\,\Delta t,$$

with heading from integrated gyro or magnetometer. Errors in v and ψ integrate directly into position; a constant heading error $\delta\psi$ produces a cross-track error growing as $v\,t\,\delta\psi$ — linear in both distance and heading error.

```
import numpy as np
def dead_reckon_2d(speed, heading, dt, p0=(0.,0.)):
    x, y = p0; traj=[(x,y)]
    for v, psi in zip(speed, heading):
        x += v*np.cos(psi)*dt; y += v*np.sin(psi)*dt
        traj.append((x,y))
    return np.array(traj)
```

In Trinetra-AI, dead reckoning is the high-rate "predict" step between absolute fixes: the IMU dead-reckons pose continuously, and each magnetic-map or GPS fix corrects the accumulated DR error inside the Kalman filter. DR alone fails (drift $\propto t^2$–t^3), so it is never used alone — but it is what keeps the estimate alive *between* fixes, which in a GPS-denied setting may be many seconds apart.

Part IV — GPS-Denied Absolute Navigation

7 Magnetic Navigation

7.1 Earth's field and the geomagnetic model

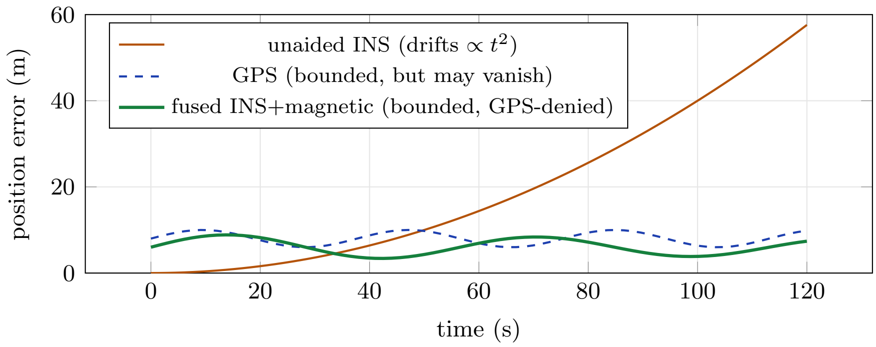
Magnetic navigation uses Earth's magnetic field for positioning and heading. The field vector has **intensity** ($\sim$25–65 μT), **declination** δ (angle from true north), and **inclination** (dip). The smooth main field is described by the **World Magnetic Model (WMM)** / IGRF; superimposed **crustal anomalies** (from iron-rich rock) form a stable spatial *fingerprint* — the World Digital Magnetic Anomaly Map (WDMAM).

7.2 Heading, mapping, and matching

Heading (tilt-compensated, from the horizontal field with roll ϕ, pitch θ from the accelerometer): $\psi_{\text{true}} = \text{atan2}(-m_y^h, m_x^h) + \delta$. **Positioning** (map-matching): treat the pre-surveyed anomaly field $B_{\text{map}}(\mathbf{p})$ as a lookup; the measurement likelihood at candidate position $\mathbf{p}$ is

$$p(\mathbf{z} \mid \mathbf{p}) \propto \exp\Big(-\tfrac{(z - B_{\text{map}}(\mathbf{p}))^2}{2\sigma^2}\Big),$$

and a Bayesian filter (a particle filter, §8) fuses this with inertial motion to localise against the map — no satellites required.



The picture is the whole argument: unaided INS diverges; an absolute anchor (GPS *or* magnetic map) bounds the error — and the magnetic anchor keeps working when GPS does not.

Why magnetic navigation is the future of GPS-denied systems. It is *passive* (emits nothing), *unjammable*, *available everywhere* (indoors, underwater, in space near Earth), and *drift-free* (tied to a fixed map). This is exactly the principle Boeing flew GPS-free for hours in 2024 and that SandboxAQ (AQNav) and Q-CTRL (Ironstone Opal) commercialise with **quantum magnetometers** reading nanotesla anomalies. In Trinetra-AI the magnetometer is the primary drift anchor, and the quantum magnetometer is the future drop-in that sharpens it. The main challenge — the platform's own magnetic signature swamping the anomaly — is where AI de-noising earns its place.

8 Map Matching

Map matching aligns noisy position/measurement data to a known map to correct drift and resolve ambiguity. Variants: **road** map-matching (snap GPS to roads), **magnetic** map-matching (match field to anomaly map), **terrain** matching (TERCOM — match altitude profile to a terrain map), and **vision** map-matching (match features to a visual map). It is probabilistic because both measurement and map are uncertain.

Map matching is naturally a **Hidden Markov Model**: hidden states are candidate positions/segments, the **emission** $p(\mathbf{z}_t \mid \mathbf{x}_t)$ scores how well a measurement fits a location, and the **transition** $p(\mathbf{x}_t \mid \mathbf{x}_{t-1})$ encodes motion feasibility. The most likely path is decoded by **Viterbi** (Newson & Krumm, 2009). For continuous fields, a **particle filter** implements the same Bayes recursion: each particle is a candidate position, weighted by the emission likelihood $\exp(-(z - B_{\text{map}})^2/2\sigma^2)$, propagated by inertial motion, and resampled — which naturally handles the *multi-modality* (several places share a field value) that a Kalman filter cannot.

```
import numpy as np
def magnetic_mapmatch_step(particles, w, imu_delta, z, B_map, sigma=1.0):
    particles = particles + imu_delta + np.random.randn(*particles.shape)*0.3 # predict
    pred = np.array([B_map(p) for p in particles]) # map lookup
    w = w * np.exp(-0.5*((z - pred)/sigma)**2); w /= w.sum() # emission
    est = np.average(particles, axis=0, weights=w)
    if 1.0/np.sum(w**2) < 0.5*len(w): # resample
        idx = np.random.choice(len(w), len(w), p=w)
        particles, w = particles[idx], np.ones(len(w))/len(w)
    return particles, w, est
```

Trinetra-AI uses **magnetic map-matching via a particle filter** as its main GPS-denied absolute-positioning method: inertial dead reckoning propagates the particles, the measured field weights them against a pre-surveyed anomaly map, and the resulting position fix is fed to the EKF/UKF core to reset drift — the software realisation of quantum magnetic navigation.

9 Localization

Localization is estimating the pose (position + orientation) within a known map or frame. **Global** localization solves the "kidnapped robot" problem (no prior guess, possibly multi-modal); **local** localization (pose tracking) refines a known pose. **Pose** = position + orientation; the estimate carries a **covariance** that quantifies uncertainty.

MCL (Dellaert, Fox, Burgard & Thrun, 1999) is a particle filter over pose. Each particle is a candidate pose; the recursion is

$$\text{predict: } \mathbf{x}_t^{(i)} \sim p(\mathbf{x}_t \mid \mathbf{x}_{t-1}^{(i)}, \mathbf{u}_t), \qquad \text{update: } w^{(i)} \propto p(\mathbf{z}_t \mid \mathbf{x}_t^{(i)}, \text{map}),$$

followed by resampling. It represents *arbitrary, multi-modal* beliefs — essential for global localization and magnetic map-matching, where several poses may explain the data until motion disambiguates them.

Bayesian localization is the general framework (predict/update on the pose posterior); MCL is its sample-based implementation, the EKF its Gaussian one. Trinetra-AI uses the Gaussian EKF/UKF for smooth pose tracking and a particle filter for the multi-modal magnetic map-matching sub-problem.

Localization is the payoff of the whole stack: fused sensors → a pose estimate with honest covariance, anchored to a magnetic/terrain map when GPS is gone. The covariance is also the *navigation confidence* the system reports — critical for safety in a GPS-denied autonomous platform.

Part V — Integration & Intelligence

10 Path Planning

Path planning finds a feasible, ideally optimal, route from start to goal. **Global** planning uses a known map (search a graph); **local** planning reacts to immediate obstacles. **Motion/trajectory planning** adds dynamics and timing. Localization answers "where am I?"; planning answers "how do I get there?".

A* (Hart, Nilsson & Raphael, 1968) searches a graph by expanding the node minimising

$$f(n) = g(n) + h(n),$$

where $g(n)$ is the cost so far and $h(n)$ is an *admissible* heuristic (never overestimates) estimating cost-to-go. With $h \equiv 0$ it reduces to **Dijkstra**. **RRT** (LaValle, 1998) and **RRT*** (Karaman & Frazzoli, 2011) grow random trees for high-dimensional/continuous spaces (RRT* is asymptotically optimal); **PRM** (Kavraki et al., 1996) builds a roadmap by sampling; **potential fields** treat the goal as attractive and obstacles as repulsive; **Hybrid A*** respects vehicle kinematics.

Planner	Idea	Use
Dijkstra	uniform-cost search	optimal on graphs, no heuristic
A*	heuristic-guided search	optimal + fast on grids
RRT / RRT*	random tree growth	high-dim / continuous; RRT* optimal
PRM	sampled roadmap	multi-query planning
Potential fields	attract/repel forces	fast local avoidance (local minima risk)
Hybrid A*	kinematic A*	car-like vehicles

Is path planning required in Trinetra-AI? Trinetra-AI is primarily a *state-estimation / localization* framework — its core deliverable is knowing *where you are* when GPS is gone, not steering. Path planning is therefore *optional / downstream*: it consumes the localization output but is not part of the estimation contribution. Include a simple A* module only if the demo needs autonomous motion; otherwise treat planning as future integration and keep the thesis focused on the fusion/navigation estimate.

11 Navigation Algorithms

The navigation *modes* — combinations of sensors and estimators — with their trade-offs.

Mode	Principle	Drift	GPS-denied?
INS (inertial only)	integrate IMU	unbounded (t^3)	yes (but drifts)
GPS only	trilateration	none	no (needs sky)
INS + GPS (loose/tight)	fuse inertial + fixes	bounded when GPS present	partial
Magnetic navigation	field/anomaly matching	bounded (map)	**yes**
Magnetic-Inertial (MINS)	INS + mag map-match	bounded	**yes**
Visual navigation	camera features	bounded (with loop closure)	yes (needs light/-texture)
Visual-Inertial (VIN/VIO)	camera + IMU	low, bounded	yes
Quantum navigation	quantum-mag + INS + AI	very low, bounded	**yes (frontier)**

Trinetra-AI's mode is Magnetic-Inertial navigation with AI, extensible to visual-inertial and quantum channels. The INS provides the continuous backbone; magnetic map-matching provides the GPS-denied absolute anchor; the barometer bounds altitude; GPS resets drift when present; and the AI layer adapts the fusion. Visual-inertial (camera/LiDAR) and the quantum magnetometer are the planned future channels — all plugging into the same fusion core.

12 AI in Navigation

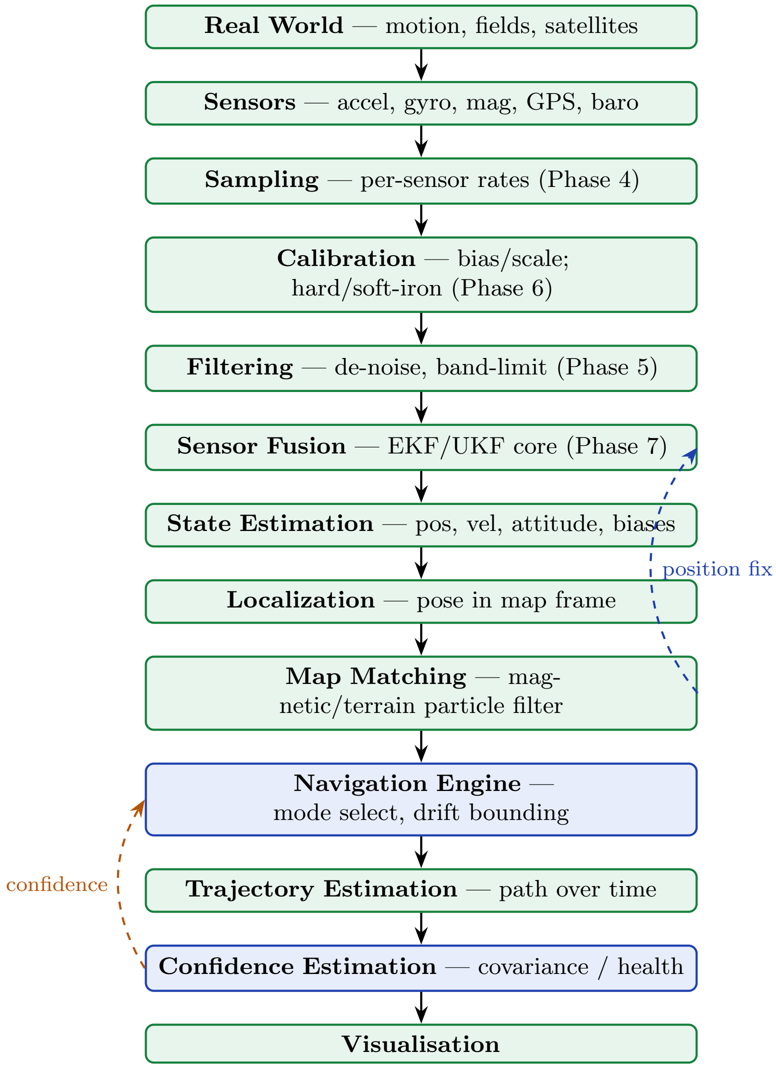
Classical navigation is optimal only under assumptions real environments break. AI relaxes them — the core research contribution of Trinetra-AI.

AI role	What it does	Best-fit model
Localization	learned measurement/observation models	CNN/Transformer
Trajectory prediction	forecast future pose	LSTM / Transformer
Drift prediction	forecast & pre-correct bias	LSTM / TCN
Sensor health	detect degradation	autoencoder / classifier
Map matching	learned field/feature matching	CNN / GNN
Adaptive weighting	set per-sensor $\mathbf{Q}, \mathbf{R}$	attention network
Fault detection	flag bad measurements	one-class / autoencoder
Navigation confidence	calibrated uncertainty	Bayesian/ensemble heads

Paradigm	Where it helps navigation
LSTM / TCN	sequential drift/trajectory forecasting from IMU windows (RoNIN/TLIO)
Transformers	long-range multi-sensor fusion; learned Kalman gains; MagNav denoising
GNNs	relational fusion of heterogeneous sensors / map graphs
Reinforcement learning	*policy* decisions: when to trust/switch sensors, active sensing
GANs	augment rare maneuvers; synthesise sensor data; anomaly via reconstruction
Diffusion models	probabilistic trajectory generation & imputation of missing data

Where AI fits in Trinetra-AI: the AI modules wrap the classical navigation core, not replace it. They forecast and pre-correct drift, de-noise the magnetic signal (the key MagNav challenge), set adaptive fusion weights and confidence, detect faults, and predict trajectory — while the Kalman/particle core keeps the interpretability and guarantees. Transformers and neural-Kalman (KalmanNet-style) are the strongest research bets; GANs/VAEs serve augmentation and anomaly detection; RL is for higher-level sensor-management policy.

13 Navigation Pipeline inside Trinetra-AI



Block detail. Sensors are sampled, calibrated, and filtered (Phases 4–6); the *fusion* core (Phase 7) produces the state; *localization* places it in the map frame; *map matching* supplies the GPS-denied absolute fix that resets drift (dashed loop back to fusion); the *navigation engine* selects the mode (GPS present vs denied) and bounds drift; *confidence estimation* reports covariance/health for safety. This is the navigation realisation of the fusion engine built in Phase 7.

Part VI — Resources, Practice & Research

14 Navigation Datasets

Real, verified; confirm licence/download on the official page. Difficulty is relative to a solo project.

Dataset	Sensors	Ground truth	Diff.	Best for
KITTI	IMU+GPS/RTK+LiDAR+cam	RTK-GPS/INS	Med	driving nav & odometry (Geiger et al.)
EuRoC MAV	IMU + stereo	Vicon/Leica	Med	drone VIO (Burri et al., 2016)
TUM VI	IMU + stereo	partial MoCap	Med	visual-inertial (Schubert et al., 2018)
RoNIN	IMU (phone)	3D traj.	Med	neural inertial nav (Herath et al., 2020)
TLIO	IMU (headset)	VIO	Med	learned inertial (Liu et al., 2020)
OxIOD	IMU (accel/gyro/mag)	Vicon/VI	Easy–Med	inertial odometry (Chen et al.)
comma2k19	IMU+GNSS+CAN	GNSS+fusion	Med	GPS/INS driving fusion
KAIST Urban	IMU+GPS+LiDAR	fused	Hard	urban-canyon GNSS-degraded
MagPIE	IMU + magnetometer	cm-level	Med–Hard	magnetic map-matching (Hanley et al., 2017)
DAF-MIT AIA MagNav	aeromagnetic + INS	flight ref.	Hard	magnetic/quantum navigation (Gnadt et al., 2023)

Recommended: KITTI or EuRoC to validate the INS/fusion navigation core against strong ground truth; MagPIE + DAF-MIT AIA for the magnetic map-matching (GPS-denied / quantum) path. Simulate a NavIC/GPS outage by masking the GNSS channel to test GPS-denied behaviour honestly.

15 Research Papers

Verified, correctly attributed. Priority: ⋆⋆⋆ read first, ⋆⋆ core, ⋆ as needed.

15.1 Foundational — navigation & estimation

Work	Authors, year	Contribution	Pri.
A New Approach to Linear Filtering & Prediction	Kalman, 1960	the Kalman filter	⋆⋆⋆
Strapdown Inertial Navigation Technology	Titterton & Weston, 2004	INS reference	⋆⋆⋆
Principles of GNSS, Inertial & Multisensor Navigation	Groves, 2013	GNSS/INS integration bible	⋆⋆⋆
Global Positioning System: Signals, Measurements & Performance	Misra & Enge, 2006	GPS reference	⋆⋆
Probabilistic Robotics	Thrun, Burgard & Fox, 2005	localization/SLAM/MCL	⋆⋆⋆
Monte Carlo Localization for Mobile Robots	Dellaert, Fox, Burgard & Thrun, ICRA 1999	particle-filter localization	⋆⋆
Hidden Markov Map Matching Through Noise & Sparseness	Newson & Krumm, ACM GIS 2009	HMM map matching	⋆⋆

15.2 Magnetic / quantum / learned navigation (most relevant)

Paper	Authors, year	Helps Trinetra-AI	Pri.
DAF-MIT AIA Open Flight Data for MagNav	Gnadt et al., 2023	magnetic-navigation benchmark & data	⋆⋆⋆
Quantum-assured magnetic navigation	Muradoglu et al., 2025 (arXiv:2504.08167)	quantum-mag nav vs strategic INS	⋆⋆
MagPIE magnetic positioning dataset	Hanley et al., IPIN 2017	indoor magnetic map-matching	⋆⋆
RoNIN / TLIO / IONet	Herath 2020 / Liu 2020 / Chen 2018	learned inertial navigation	⋆⋆⋆
Denoising IMU Gyroscopes with DL	Brossard et al., RA-L 2020	learned de-noising for MagNav/INS	⋆⋆
KalmanNet: NN-aided Kalman filtering	Revach et al., IEEE TSP 2022	learned-gain adaptive navigation	⋆⋆
VINS-Mono / ORB-SLAM	Qin et al., T-RO 2018 / Mur-Artal et al., 2015	visual-inertial & SLAM references	⋆

Reading path: Groves (GNSS/INS integration) → Titterton & Weston (INS) → Thrun (localization) for the classical core; then Gnadt (MagNav data) → Muradoglu (quantum) → RoNIN/TLIO (learned inertial) for the frontier Trinetra-AI targets. **Citation hygiene:** verify every DOI/venue on the official page; cite the NavIC crisis and quantum-nav industry milestones (Boeing 2024, SandboxAQ, Q-CTRL) from primary sources / official statements.

16 Practical Implementation

Notebook	Contents	Expected output
`01_gps_solver.ipynb`	pseudorange least squares; NMEA parse	position fix + DOP
`02_coord_transforms.ipynb`	geodetic/ECEF/ENU/NED	verified transforms
`03_ins_deadreckon.ipynb`	strapdown mechanization	drifting INS trajectory
`04_ins_gps_ekf.ipynb`	loosely-coupled INS/GPS EKF	bounded fused trajectory
`05_magnetic_heading.ipynb`	tilt-comp. heading + declination	heading vs truth
`06_mag_mapmatch_pf.ipynb`	particle-filter map matching	GPS-denied position fix
`07_localization_mcl.ipynb`	Monte Carlo localization	pose posterior
`08_nav_dashboard.ipynb`	full pipeline + metrics	trajectory + confidence dashboard

Evaluation metrics. ATE/RTE (trajectory error), position RMSE, heading error (deg), *drift rate* in GPS-denied segments (m per 100 m), fix availability, and filter consistency (NEES/NIS). **Visualisation:** trajectory vs ground truth on a map, covariance ellipses, error-vs-time (with GPS-outage windows shaded), magnetic-map overlays.

Debugging. Diverging estimate ⇒ check frame conventions (ENU/NED sign bugs), units (deg/rad, $g/\mathrm{m\,s^{-2}}$), time sync, and $\mathbf{Q}, \mathbf{R}$ scaling; a biased innovation signals a model/calibration error. **Best practices:** validate each mode against a dead-reckoning-only and a GPS-only baseline; simulate GPS outages explicitly; unit-test coordinate transforms; report confidence honestly (NEES). **Common mistakes:** ENU/NED confusion; forgetting declination; not gating GPS multipath; ignoring the platform's own magnetic signature in map-matching; evaluating only *with* GPS (hiding the GPS-denied weakness).

17 Research Contribution

Research gaps & M.Tech-thesis-scale contributions: (1) *AI-de-noised magnetic map-matching*: learn to strip the platform's magnetic signature (the core MagNav problem) and quantify the resulting GPS-denied positioning gain — directly building on the DAF-MIT AIA data. (2) *Learned adaptive navigation*: a network setting fusion mode and $\mathbf{Q}, \mathbf{R}$ from context (GPS quality, magnetic richness), benchmarked against fixed-gain fusion. (3) *Neural-Kalman navigation*: a KalmanNet-style learned gain for magnetic-inertial navigation under model mismatch. (4) *Confidence-aware GPS-denied navigation*: calibrated uncertainty that triggers graceful mode-switching and safe fallback.

Future directions. *Quantum navigation* — integrate a (simulated then real) quantum-magnetometer channel and quantify drift reduction. *Magnetic navigation improvements* — richer anomaly maps, learned matching, robustness to temporal field variation. *Adaptive/neural navigation* — end-to-end learned components wrapped around a guaranteed classical core. A defensible thesis centres on **AI-enhanced magnetic-inertial navigation for GPS-denied environments**, with rigorous classical baselines, explicit GPS-outage evaluation, and ablations isolating each AI module.

18 Interview & Research Preparation

18.1 60 interview questions

1. Define navigation / PNT.
2. Smooth-vs-absolute source tension.
3. Why is navigation hard?
4. GPS three segments.
5. What is a pseudorange?
6. Why ≥ 4 satellites?
7. Trilateration vs triangulation.
8. What is DOP?
9. Carrier phase vs code.
10. RTK positioning.
11. Multipath and why it breaks a KF.
12. Ionospheric delay.
13. Why GPS fails indoors.
14. Jamming vs spoofing.
15. GPS vs GNSS.
16. Name six constellations.
17. What is NavIC and its coverage?
18. Why did NavIC fall below threshold?
19. Multi-GNSS benefits.
20. Why GNSS still fails GPS-denied.
21. What is an INS?

22. Strapdown vs gimballed.
23. INS mechanization equations.
24. Why INS drifts $\propto t^3$.
25. INS grades.
26. Why INS is essential GPS-denied.
27. Dead reckoning concept.
28. 2-D DR equations.
29. Heading-error cross-track growth.
30. Why DR alone fails.
31. Earth field components.
32. Declination vs inclination.
33. WMM vs crustal anomalies.
34. Tilt-compensated heading.
35. Magnetic map-matching principle.
36. Why magnetic nav is unjammable.
37. Quantum magnetometer role.
38. What is map matching?
39. Road vs magnetic vs terrain matching.
40. HMM map matching.
41. Why a particle filter for map matching?
42. Global vs local localization.
43. Monte Carlo Localization.
44. Kidnapped-robot problem.
45. Pose & covariance.
46. Path planning vs localization.
47. A* $f = g + h$.
48. Admissible heuristic.
49. Dijkstra vs A*.
50. RRT vs RRT*.
51. Is planning needed in Trinetra-AI?
52. Geodetic vs ECEF vs ENU.
53. ENU vs NED.
54. INS+GPS loose vs tight.
55. Visual-inertial navigation.
56. Magnetic-inertial navigation.
57. Where LSTM helps navigation.
58. Where transformers help.
59. Where RL helps.
60. Navigation confidence.

18.2 40 viva questions

1. Derive the pseudorange solution.
2. Explain DOP geometry.
3. Derive INS mechanization.
4. Show INS error growth orders.
5. Derive 2-D dead reckoning error.
6. Derive tilt-compensated heading.
7. Set up magnetic map-matching likelihood.
8. Formulate map matching as an HMM.
9. Explain Viterbi decoding.
10. Derive the MCL recursion.
11. Why particle filter over EKF for map matching?
12. Derive geodetic→ECEF.
13. ECEF→ENU rotation.
14. ENU/NED pitfalls.
15. Loosely vs tightly coupled INS/GPS.
16. Choose $\mathbf{Q}, \mathbf{R}$ for INS/GPS.
17. Gate GPS multipath.
18. A* optimality & admissibility.
19. RRT* asymptotic optimality.
20. When is planning unnecessary here?
21. Bound INS drift with an absolute anchor.
22. Explain the error-vs-time figure.
23. NavIC crisis & its lesson.
24. Why multi-GNSS doesn't solve denial.
25. Magnetic signature de-noising challenge.
26. Quantum-magnetometer advantage.
27. Confidence from covariance.
28. NEES/NIS consistency.
29. Mode-switching logic (GPS ↔ denied).
30. Terrain-referenced navigation.
31. Visual-inertial observability.
32. Loop closure.
33. Where does AI enter the pipeline?
34. KalmanNet for navigation.
35. Evaluate GPS-denied honestly.
36. Simulate a GPS outage.
37. Baselines for a nav claim.
38. Ablate a nav module.
39. Failure modes of learned navigation.
40. Explain the full pipeline.

18.3 30 research questions

1. Learned vs classical magnetic de-noising: positioning gain?
2. Adaptive fusion-mode selection by AI.
3. Neural-Kalman for magnetic-inertial nav.
4. Drift-reduction ceiling from a quantum magnetometer.
5. Robustness of map-matching to field temporal variation.
6. Calibrated confidence for safe mode-switching.
7. Cross-platform MagNav generalisation.
8. Learned map-matching vs particle filter.
9. Tight vs loose coupling under denial.
10. Observability with intermittent aiding.
11. Transformer fusion over long outages.
12. RL for active sensor management.
13. GAN augmentation for rare trajectories.
14. Diffusion trajectory imputation.
15. Edge-deployable navigation on Jetson.
16. Online map building (magnetic SLAM).
17. Anomaly-map resolution vs accuracy.
18. Fault-robust navigation under sensor failure.
19. Sim-to-real gap for learned nav.
20. Multi-GNSS + INS + mag optimal weighting.
21. Quantum-noise modelling in nav fusion.
22. Benchmark design for GPS-denied navigation.
23. NavIC-outage resilience study.
24. Confidence calibration (reliability diagrams).
25. Long-horizon trajectory forecasting.
26. Terrain + magnetic multi-map fusion.
27. Data efficiency of hybrid vs end-to-end nav.
28. Loop closure in magnetic navigation.
29. Uncertainty propagation through learned modules.
30. Graceful degradation guarantees.

18.4 20 coding questions

1. Solve GPS position from pseudoranges.
2. Parse NMEA to lat/lon/alt.
3. Compute DOP from geometry.
4. Geodetic↔ECEF↔ENU.
5. Strapdown INS dead reckoning.
6. Loosely-coupled INS/GPS EKF.

7. Tilt-compensated magnetic heading.
8. Apply WMM declination.
9. Particle-filter magnetic map-matching.
10. Monte Carlo Localization on a grid.
11. HMM map matching + Viterbi.
12. 2-D dead reckoning with heading.
13. A* on a grid map.
14. RRT in continuous space.
15. Chi-square GPS gate.
16. Simulate a GPS outage & measure drift.
17. Plot trajectory vs ground truth + covariance.
18. Compute ATE/RTE.
19. Compute NEES over a run.
20. Build the navigation confidence signal.

19 Summary, Glossary & Roadmap

19.1 Summary

Navigation estimates position, velocity, orientation, heading, and trajectory — reliably, even without GPS. GPS/GNSS give absolute fixes but fail indoors, underwater, and under jamming (NavIC's 2024–26 crisis makes the fragility concrete). The INS is self-contained but drifts without bound. The resolution is fusion plus *absolute anchors*: magnetic-map matching (passive, unjammable, drift-free), barometric altitude, and GPS when present, all combined in the Kalman/-particle core from Phase 7. Map matching (HMM / particle filter) and localization (MCL / Bayesian) place the estimate in a map frame; AI wraps the core to de-noise, adapt, predict drift, and report confidence. Path planning is downstream and optional. This phase turns the fusion engine into a working GPS-denied navigation system.

19.2 Glossary

ATE/RTE — absolute/relative trajectory error. **DOP** — dilution of precision (satellite-geometry error amplification). **Dead reckoning** — pose by integrating motion from a known start. **ECEF/ENU/NED** — Earth-fixed / local tangent frames. **GNSS** — all satellite-navigation systems (GPS is one). **INS** — inertial navigation system. **MCL** — Monte Carlo (particle-filter) localization. **Map matching** — aligning measurements to a known map. **NavIC** — India's regional GNSS (IRNSS). **PNT** — position, navigation, timing. **Pseudorange** — GPS range including receiver clock bias. **WMM / anomaly map** — geomagnetic model / crustal-field fingerprint.

19.3 Roadmap: Phase 8 → AI Navigation, Quantum Navigation & the full architecture

Next	**What this phase hands it**
AI-based navigation	the hooks for learned drift/de-noising/adaptive weighting/confidence around the nav core
Quantum navigation	the magnetic map-matching pipeline into which a quantum-magnetometer channel drops directly
Full Trinetra-AI architecture	the complete GPS-denied navigation system: INS backbone + magnetic anchor + baro + AI, GPS when present

You now have a working navigation system, not just an estimator: inertial dead reckoning bounded by magnetic-map matching and barometric altitude, with GPS as an opportunistic anchor and AI adapting it all. The remaining work deepens the AI modules and integrates the quantum-magnetometer channel — the frontier that turns Trinetra-AI from a strong GPS-denied navigator into a quantum-ready one.

End of Phase 8 notes. Next: AI-based Navigation → Quantum Navigation → the complete Trinetra-AI architecture.